

Human Motion Generation from Text, Scene, and Trajectory

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Abstract

We present a diffusion model for generating human motion conditioned jointly on natural-language text, 3D scene geometry, and a target trajectory goal. Rather than routing each condition through its own cross-attention path, we represent text, scene, and goal as tokens that are concatenated onto the motion sequence and processed by a single shared self-attention transformer, with the diffusion timestep injected via AdaLN-Zero. Each condition is trained with independent classifier-free-guidance dropout, so at sampling time any subset of text, scene, and goal can be supplied, omitted, or guided at its own strength. We train on NymeriaPlus, a large-scale dataset of real-world egocentric motion capture with paired scene geometry and text annotations, using a floor-anchored joint-position motion representation. An earlier iteration of this idea anchored the local coordinate frame on the raw pelvis position and hard-inpainted the goal into the generated window; we found this broke scene and goal conditioning whenever pose changed the pelvis’s height above the floor, and describe the representation changes – anchoring on estimated floor height instead, and treating the goal as a soft, CFG-guided token like text and scene – that fixed it. We report qualitative rollouts on held-out test sequences spanning locomotion and everyday scene interactions such as cooking, using a fridge, sitting, and lying down, and leave a quantitative evaluation to future work.

1. Introduction

Generating realistic, controllable human motion is a core requirement for embodied AI, AR/VR content creation, and human-robot interaction: an agent that must act convincingly in a 3D environment needs motion that is simultaneously plausible under physics and human biomechanics, responsive to natural-language instructions, aware of the surrounding scene geometry (it should not walk through furniture), and able to reach a specified goal location. Most prior work addresses these conditions in isolation, or combines a subset of them with bespoke, per-modality architec-

ture – a dedicated cross-attention branch for text [10, 13], a separate module for scene or trajectory constraints [6, 11]. This raises the cost of adding a new condition and makes it harder to reason about how conditions trade off against one another at generation time.

We instead treat text, scene, and goal as three instances of the same mechanism: each is projected into a small set of tokens and appended to the motion token sequence, so a single shared self-attention layer performs all cross-modal reasoning between motion and conditions, with no modality-specific attention path. The diffusion timestep does not enter this token sequence either; every transformer block is modulated by AdaLN-Zero instead. Each condition modality is independently dropped out during training in the style of classifier-free guidance (CFG), so at inference time the model has, by construction, been trained to generate from any subset of {text, scene, goal} and to be steered toward any one of them more or less strongly via its own guidance scale.

We train and evaluate this architecture on NymeriaPlus, a large-scale dataset of real-world egocentric human motion capture with paired 3D scene geometry and atomic text annotations (Sec. 3). Over the course of this project the motion representation itself changed in a way that turned out to matter as much as the architecture: an early iteration anchored each window’s local coordinate frame on the raw pelvis position, hard-inpainted the goal into the generated window, and generated 50-frame windows; we found this both made scene conditioning ineffective (the pelvis height drifts with pose – standing, sitting, and lying down all put the pelvis at different heights above the same floor, corrupting the local frame the scene crop and goal were expressed in) and made the goal unable to be traded off against text or scene. Re-anchoring the local frame on an estimated floor height, and turning the goal into a soft, CFG-guided token exactly like text and scene, fixed both problems.

We report qualitative rollouts on held-out test-split sequences with the current (floor-anchored) checkpoint, spanning locomotion and scene-interaction behaviors (Sec. 4); quantitative evaluation against ground truth is left as future work (Sec. 5).



Figure 1. Autoregressive rollouts on held-out test-split segments, conditioned jointly on text, scene, and goal. Each row shows eight evenly-sampled frames in temporal order (left to right) from one continuous rollout, with the conditioning text prompt below. Full animated rollouts are in the project repository’s README.

In summary, our contributions are:

- A token-concatenation architecture for jointly conditioning motion diffusion on text, scene, and goal through one shared self-attention, with independent per-modality CFG dropout.
- A floor-anchored, soft-goal-token motion representation that fixes scene- and goal-conditioning failure modes present in an earlier, pelvis-anchored, hard-inpainted-goal design.

- Qualitative results on NymeriaPlus’s held-out test split, demonstrating scene- and goal-appropriate motion across a range of behaviors.

2. Related Work

Motion diffusion models. Denoising diffusion has become a standard generative model for human motion, typically predicting clean motion (rather than noise) directly and denoising a fixed-length window in place, as in

MDM [10] and MotionDiffuse [13]. We follow this clean-motion-prediction formulation, with a cosine noise schedule and a transformer denoiser.

Text-conditioned motion generation. A large body of work generates motion from natural-language descriptions, most commonly by cross-attending motion tokens to frozen or fine-tuned language-model features – TEMOS [7] aligns a motion VAE to a text encoder, MotionCLIP [9] aligns motion to CLIP’s latent space directly, and T2M-GPT [12] autoregresses over a discrete VQ-VAE motion vocabulary conditioned on text. We instead concatenate text tokens onto the motion sequence and let one shared self-attention layer handle the interaction, which is what lets us add scene and goal conditioning the same way without new architecture.

Scene- and goal-conditioned motion. Conditioning motion synthesis on 3D scene geometry or a target location/trajectory is typically handled with a dedicated scene encoder and either cross-attention or inpainting/guidance toward the goal: HUMANISE [11] generates scene-conditioned motion from language by pairing captured motion with synthetic indoor scenes, SAMP [2] models stochastic, style-diverse scene-interaction motion toward a target object, GMD [6] and PriorMDM [8] steer a pretrained motion diffusion model toward spatial trajectory constraints through guidance and inpainting respectively, and TRUMANS [5] and LINGO [4] generate long, autoregressive human-scene interaction sequences from text and a goal on large synthetic-scene motion-capture datasets. Our goal token is deliberately *soft*: it is trained and sampled with the same CFG-dropout and guidance-scale mechanism as text and scene, rather than hard-inpainted into the generated window, so the model can trade it off against what the text or scene imply instead of being forced to reach it regardless.

Table 1 summarizes how these methods compare along the axes that matter for our setting: which conditions they support, whether they train on real or synthetic motion-scene capture, and how much motion data that training draws on. HUMANISE and MDM each support two of the three conditions we target (scene and text without trajectory, and text without scene, respectively); TRUMANS and LINGO support all three but, like HUMANISE, train on synthetic scenes rather than real-world capture. NymeriaPlus (Sec. 3.1) is what makes training on all three conditions with real motion-scene capture practical in the first place – even after the filtering in Sec. 3.1, our training corpus dwarfs every dataset in the table, real or synthetic, and is the only one at this scale to pair real, in-the-wild motion with both text and scene annotation. (NymeriaPlus’s raw, unfiltered capture is wider still, at roughly 300 hours.)

Table 1. Prior work compared on conditioning support, data source, and training-motion volume (methods above the rule, the datasets they draw on below it). NymeriaPlus is the only dataset pairing real (not synthetic-scene) capture with all three conditions, on by far the largest amount of motion data.

Method	Data	Text	Scene	Traj.	Hours
HUMANISE [11]	Synthetic	✓	✓	–	11
MDM [10]	Real	✓	–	–	–
TRUMANS [5]	Synthetic	–	✓	✓	15
LINGO [4]	Synthetic	✓	✓	✓	16
HumanML3D [1]	Real	✓	–	–	28.6
NymeriaPlus	Real	✓	✓	✓	182

Classifier-free guidance. CFG [3] jointly trains a conditional and an unconditional model and combines their score estimates at sampling time, avoiding the separate classifier that earlier guidance methods required. It is normally applied to a single condition; we apply it independently per modality (text, scene, goal), each with its own dropout probability during training and its own guidance scale at sampling time, so any present modality can be guided toward or dropped without affecting the others.

3. Method

3.1. Dataset

We use NymeriaPlus, a large-scale multimodal dataset of real-world human motion captured with egocentric multi-camera video, IMU, SMPL body tracking, 3D scene meshes, and atomic-action text annotations. After filtering for sequences with complete SMPL tracking, object geometry, and narrations, and discarding annotations shorter than roughly one second or longer than 60 frames, the usable corpus is 707 sequences, 6,550,431 frames (181.96 hours at 10 FPS), and 144,356 atomic text annotations. The 707 sequences are split at the *sequence* level – an entire recording, participant, and room, with every one of its annotations, goes to exactly one split – into 548 train, 83 validation, and 76 test sequences, so test-split rooms and motions are never seen during training. Environment geometry is voxelized into a boolean 3D occupancy grid at 5 cm resolution per sequence.

3.2. Motion Representation

Each frame is represented as 24 SMPL joint positions in metric 3D space (no rotations), flattened to a 72-dimensional feature. Every window a model sees – its trailing history frames, the frames being generated, the goal, and the scene crop – is expressed relative to one fixed *anchor*: the world-space position and facing yaw of the last history frame (or the window’s own first frame when there is no history). The anchor’s horizontal position is the pelvis (x, y) ,

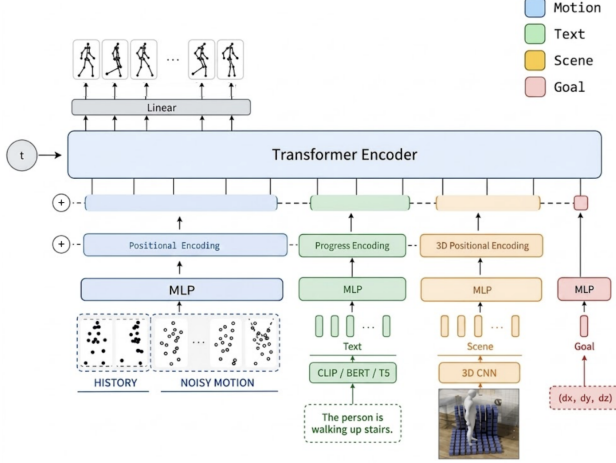


Figure 2. Motion, text, scene, and goal all enter one transformer as tokens in a single shared self-attention sequence; the diffusion timestep modulates every block via AdaLN-Zero instead of occupying a token.

but its vertical component is an *estimated floor height* at that instant rather than the pelvis’s own height. This distinction matters: pelvis height drifts by tens of centimeters between standing, sitting, and lying down, so anchoring vertically on the pelvis makes the local frame – and everything expressed in it, including the scene crop and the goal – drift with pose rather than staying tied to the physical ground.

3.3. Architecture

We use an 8-layer, 512-dimensional transformer denoiser (8 attention heads, 1024-dimensional feed-forward blocks) that predicts clean motion directly from a noised window, following a cosine-schedule DDPM with 50 timesteps. History frames are fixed (re-inpainted) at every denoising step; the loss is a position MSE over the non-history frames only.

Shared self-attention, no cross-attention. As shown in Fig. 2, every enabled condition – text, scene, goal – is turned into a small set of tokens and concatenated directly onto the motion token sequence (history plus noisy future). One self-attention layer per block processes the whole concatenated sequence; there is no separate cross-attention path per modality. A learned per-modality token-type embedding and, for motion tokens, a sinusoidal positional encoding let attention distinguish history from future and one modality from another beyond content alone.

Text. Text is encoded with a frozen language model (CLIP, DistilBERT, or T5 – T5 in our trained checkpoints) and kept as a sequence of per-token embeddings, padded or truncated to a fixed token budget, rather than pooled to a single vector. This lets individual phrases in a prompt line up with individual phases of the generated window through self-attention. A window-progress scalar (where in its an-

notation’s span the sampled window falls) is embedded and added to the text encoder’s output, since a training window’s start point within its annotation is randomized but every window sharing an annotation shares the same text.

Scene. Figure 3 shows the scene path: a small 3D CNN (strided 3D convolutions, doubling channels each stage) encodes an occupancy-grid crop around the anchor down to a fixed $4 \times 4 \times 4$ volume, so every configured input resolution yields the same 64 output tokens. Each token is positionally encoded by its own anchor-local cell center through a small MLP, tying the encoding to real geometry rather than a memorized slot index.

Goal. The goal is the window’s own last-frame pelvis position, expressed as an anchor-local displacement (dx, dy, dz) – the same physical quantity as channels [0:3] of the motion feature, normalized with the same statistics. Unlike text and scene, which describe the whole window, the goal is a single, clean target the window is known to actually reach, with no explicit arrival time (the window length implies it). It enters the shared attention sequence as one more soft, CFG-guided token, exactly like text and scene, rather than being hard-inpainted into the generated window – the model is only *encouraged* toward it by the training loss, and can trade it off against what the scene or text imply.

3.4. Multi-Condition Classifier-Free Guidance

Each of text, scene, and goal is independently dropped from the condition set during training (default 20% each), plus a small joint-dropout probability (5%) of dropping all three together. A condition simply omits its tokens from the sequence when dropped or absent – there is no learned null-vector standing in for it.

At sampling time, write $\varepsilon(T, S, G)$ for the model’s denoising prediction with text, scene, and goal condition tokens T, S, G all present, and $\emptyset$ for a modality’s tokens being dropped from that evaluation. Each present modality gets its own *guidance direction*, the difference the model’s prediction makes when that modality alone is removed:

$$\Delta T = \varepsilon(T, S, G) - \varepsilon(\emptyset, S, G) \quad (1)$$

$$\Delta S = \varepsilon(T, S, G) - \varepsilon(T, \emptyset, G) \quad (2)$$

$$\Delta G = \varepsilon(T, S, G) - \varepsilon(T, S, \emptyset) \quad (3)$$

and the final guided prediction linearly combines the fully-conditional prediction with each present modality’s direction, scaled by its own guidance weight w_T, w_S, w_G :

$$\varepsilon_{\text{guided}} = \varepsilon(T, S, G) + w_T \Delta T + w_S \Delta S + w_G \Delta G. \quad (4)$$

This requires one model evaluation per present, guided modality in addition to the fully-conditional one; a guidance weight of 1.0 recovers plain conditional sampling for that modality, and a dropped modality simply contributes no term.

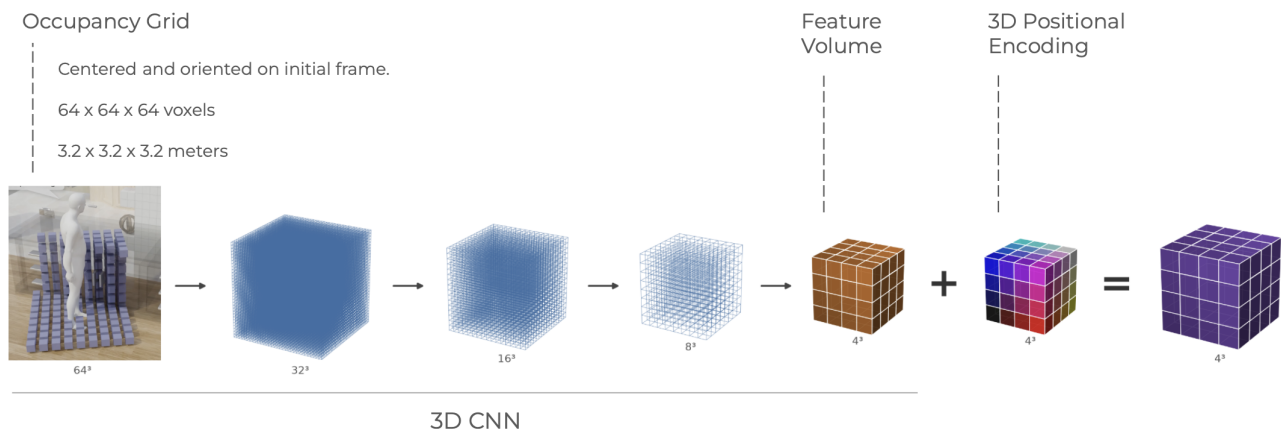


Figure 3. The scene occupancy crop around the anchor is encoded by a 3D CNN down to a fixed $4 \times 4 \times 4$ volume, giving 64 scene tokens with real 3D positional encoding.

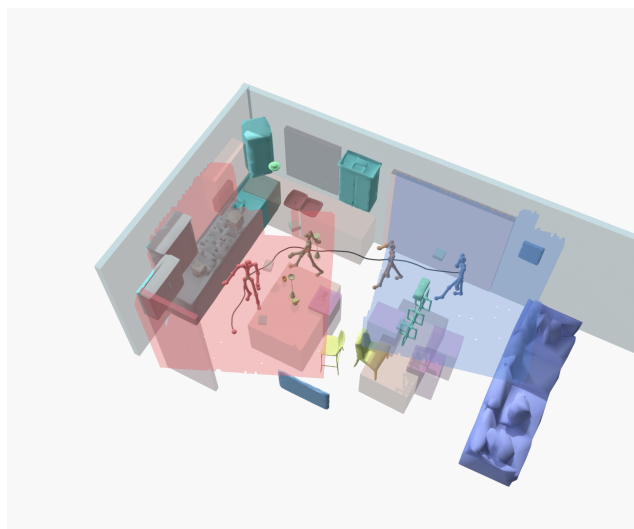


Figure 4. At inference, windows are generated autoregressively: each window’s trailing frames become the next window’s history, and the anchor – including its floor height – is re-estimated causally after every window.

3.5. Autoregressive Rollout

At inference (Fig. 4), windows are generated one at a time: a window’s trailing frames become the next window’s history, the next anchor’s horizontal position is the last generated frame’s pelvis, its facing yaw is re-estimated from the hip direction, and its floor height is re-estimated causally from history and the newly generated frames, so the anchor stays floor-relative across an arbitrarily long rollout rather than only within one window.

4. Qualitative Results

We evaluate the current architecture (Sec. 3) qualitatively, by inspecting autoregressive rollouts (Sec. 3.5) on held-out test-split segments (Sec. 3.1) – 76 sequences never seen during training or for fitting normalization statistics – conditioned jointly on text, scene, and goal. Quantitative evaluation (e.g. MPJPE against ground truth, FID, diversity, foot-skate) is left as future work (Sec. 5).

Figure 1 shows evenly-sampled frames from rollouts spanning locomotion (walking, stairs) and scene-interaction behaviors (cooking, using a fridge, sitting, lying down), each with the conditioning text prompt. In each case the generated motion respects the surrounding scene geometry (e.g. navigating around furniture on stairs, approaching and orienting toward the fridge) and produces a pose appropriate to the goal and text condition (e.g. a seated pose when sitting on the bed, a lying pose with the correct orientation on it) rather than a generic walking cycle regardless of context. These rollouts were produced with an interactive front-end that lets a user place a start position and a sequence of text- and waypoint-annotated segments directly on the scene geometry, and generates motion for the whole sequence with the guidance scales of Sec. 3.4 exposed as adjustable per-condition controls.

4.1. Guidance-Scale Ablation

To inspect what each per-modality guidance weight in Sec. 3.4 actually controls, we sample the same prompt on a synthetic scene while sweeping one condition’s guidance scale from 0 to 3 with the other two held fixed, and render the full generated trajectory (poses overlaid across time) for each sampled scale. Figure 5 shows this sweep, lowest to highest guidance left to right, for text, scene, and goal in

isolation. Increasing text guidance sharpens the described arm motion; increasing scene guidance pulls the trajectory away from the obstacle; increasing goal guidance pulls the endpoint of the trajectory toward the marked waypoints. Figure 6 isolates the comparison the scene sweep is making: its left panel is Fig. 5(b)’s highest-guidance trajectory, now placed next to the same prompt and guidance scale with the obstacle removed, showing the path bend around it only when it is actually there.

5. Limitations and Conclusion

Limitations. Our evaluation is qualitative only: we do not yet report MPJPE, FID, diversity, or foot-skate metrics against ground truth, so the improvement described in Sec. 1 (floor- rather than pelvis-anchoring, a soft rather than hard-inpainted goal token) is argued from the conditioning failure modes it fixes, not from a quantitative ablation. We also have no autoregressive-rollout error-accumulation study (Sec. 3.5 generates windows one at a time, but we only inspect the resulting rollouts visually).

Beyond evaluation, three further limitations bound how far the current system can be pushed:

- **Data.** NymeriaPlus’s scene annotations are incomplete for some sequences, its captured motion is occasionally imprecise or noisy, and dynamic objects in the scene are not represented in the (static) occupancy grid the scene encoder sees.
- **Conditions.** The goal is a strong signal relative to text and scene – because it is exactly the window’s own reachable last frame, it can dominate the other two at moderate guidance scales rather than being weighed against them as intended. Text, scene, and goal are also not independent in practice (a text instruction can imply a goal, a goal can imply a scene layout), and combining them with guidance scales outside the training distribution’s typical range can push sampling out of distribution.
- **Compute.** Training and evaluation ran on a single consumer GPU (RTX 3080, 10 GB VRAM), which bounded batch size, model scale, and the amount of hyperparameter search possible within the project’s timeline.

Conclusion. We presented a motion diffusion model that conditions jointly on text, scene, and goal by concatenating all three as tokens onto the motion sequence and processing everything through one shared self-attention transformer, with independent per-modality classifier-free-guidance dropout enabling flexible test-time conditioning (Sec. 3.4). Re-anchoring the motion representation on estimated floor height rather than raw pelvis position, and making the goal a soft CFG-guided token rather than hard-inpainting it, resolved conditioning failure modes present in an earlier iteration of the same architecture. On NymeriaPlus – whose training corpus alone is an order of mag-

nitude larger than prior real- or synthetic-scene motion datasets (Tab. 1) – we show qualitative rollouts spanning locomotion and scene-interaction behaviors that respect scene geometry and reach their conditioned goal, and a per-condition guidance-scale ablation (Sec. 4.1) demonstrating that each modality’s guidance weight controls its intended aspect of the generated motion.

Future work should prioritize three directions. The first is extending these results with quantitative metrics such as FID, R-precision, and diversity against ground truth, to replace the qualitative argument made here with a measured one. The second is investigating and improving scene conditioning specifically, since it is both the newest condition in the architecture and the one most exposed to the data limitations above. The third is exploring alternative mechanisms for combining multiple conditions beyond independent per-modality CFG, given the interdependence between text, scene, and goal noted above.

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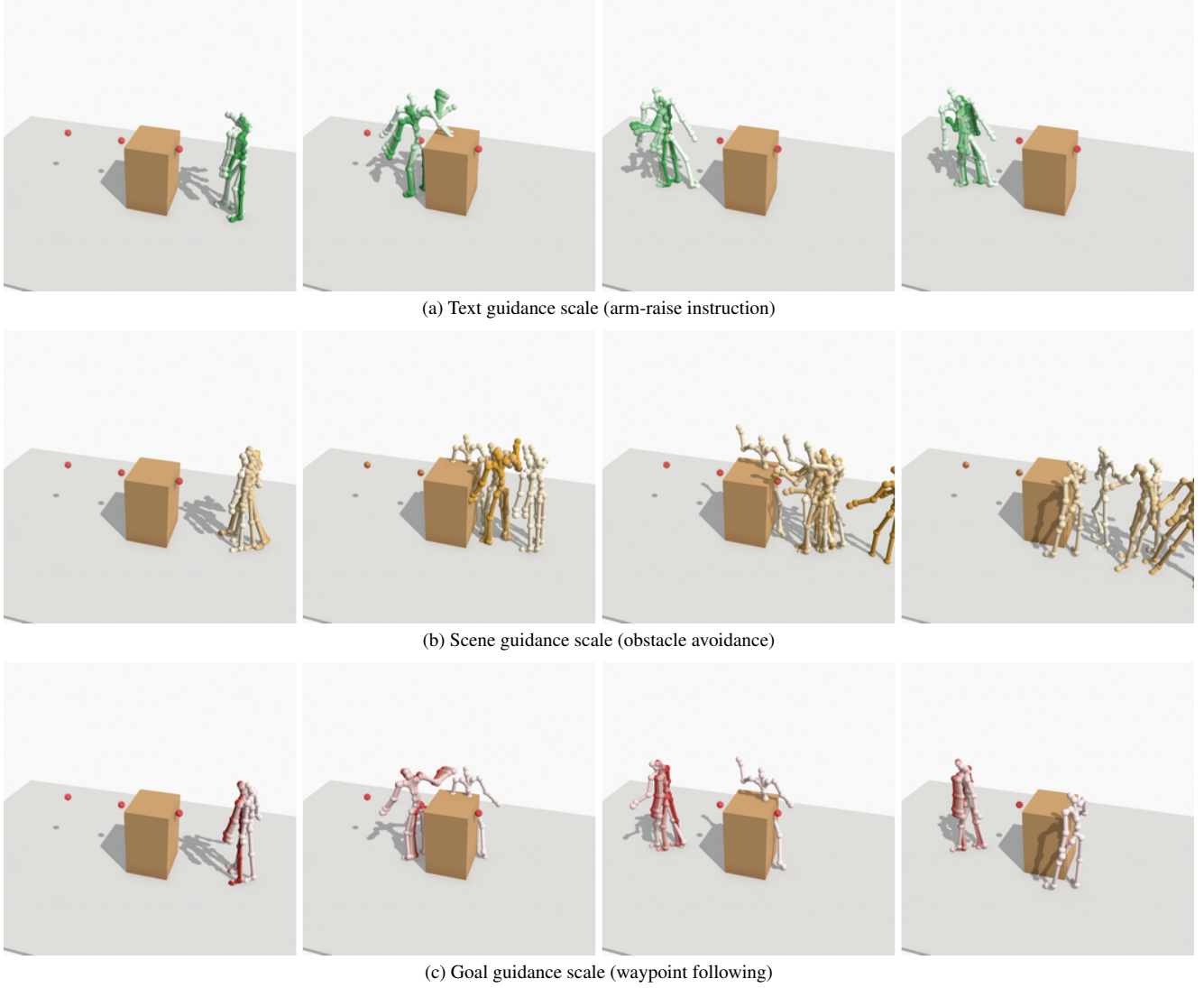


Figure 5. Per-condition classifier-free guidance sweep on a synthetic scene, one condition varied in isolation (the other two guidance scales fixed at 1.0). Each panel is the full trajectory generated at one guidance scale, sampled left to right from low to high across the 0–3 sweep.

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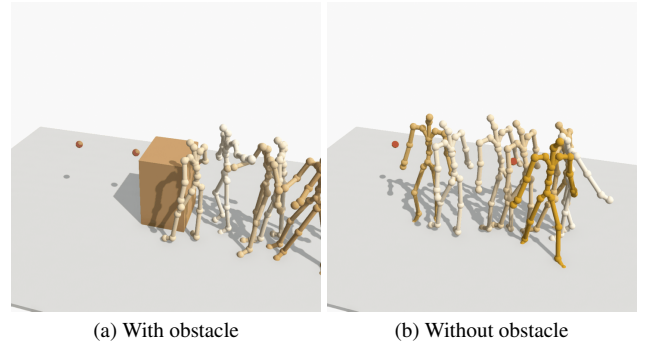


Figure 6. Highest-scene-guidance trajectory with and without the obstacle present; the generated path bends around it when present.